SQL DEVELOPER INTERNSHIP

Task 1: Database Setup and Schema Design

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Domain Selected: Digital Library Management System
Tools Used: pgAdmin / PostgreSQL

1. Database Schema SQL Script (Library Management)

The DDL script below establishes a structured relational schema for a Library Management System, implementing referential integrity, unique constraints, and check constraints.

```
-- Database creation
CREATE DATABASE library_db;

-- 1. Authors Table
CREATE TABLE authors (
    author_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    author_name VARCHAR(120) NOT NULL,
    nationality VARCHAR(50)
);

-- 2. Books Table
CREATE TABLE books (
    book_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    isbn VARCHAR(20) UNIQUE NOT NULL,
    title VARCHAR(200) NOT NULL,
    author_id INT NOT NULL,
    copies_available INT DEFAULT 1 CHECK (copies_available >= 0),
    CONSTRAINT fk_author FOREIGN KEY (author_id) REFERENCES authors(author_id) ON DELETE
RESTRICT
);

-- 3. Members Table
CREATE TABLE members (
    member_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    full_name VARCHAR(120) NOT NULL,
    phone_number VARCHAR(15) UNIQUE NOT NULL,
    membership_date DATE DEFAULT CURRENT_DATE
);

-- 4. Borrowings Table (Junction Table between Members and Books)
CREATE TABLE borrowings (
    issue_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    member_id INT NOT NULL,
    book_id INT NOT NULL,
    issue_date DATE DEFAULT CURRENT_DATE,
    return_date DATE,
    status VARCHAR(20) DEFAULT 'Issued' CHECK (status IN ('Issued', 'Returned', 'Overdue')),
    CONSTRAINT fk_member FOREIGN KEY (member_id) REFERENCES members(member_id) ON DELETE
CASCADE,
```

```
    CONSTRAINT fk_book FOREIGN KEY (book_id) REFERENCES books(book_id) ON DELETE CASCADE
);
```

2. Theoretical & Conceptual Questions

Q1. What is normalization?

> Normalization is a database design technique that decomposes tables to reduce data redundancy and eliminate operational anomalies (insert, update, and delete anomalies). It ensures that data dependencies are logical and stored efficiently.

Q2. Explain primary vs foreign key.

> - **Primary Key:** A unique attribute or set of attributes that acts as the principal identifier for every row in a table. Null values are strictly prohibited.
> - **Foreign Key:** An attribute that points to the primary key of another table to link records and enforce referential integrity across related tables.

Q3. What are constraints?

> Constraints are predefined conditions or rules declared on columns during table creation to validate incoming data and maintain data accuracy across transactions (e.g., `NOT NULL`, `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`).

Q4. What is a surrogate key?

> A surrogate key is an artificial, system-generated primary key (such as an auto-incrementing integer or UUID) that has no external meaning or business significance outside of database identification.

Q5. How do you avoid data redundancy?

> Data redundancy is prevented by designing normalized relational models (at least up to 3NF), eliminating duplicate data storage, and establishing proper Foreign Key references between entities.

Q6. What is ER diagram?

> An Entity-Relationship (ER) diagram is a conceptual diagrammatic representation of database architecture that depicts entities, their attributes, and how they relate to one another.

Q7. What are the types of relationships in DBMS?

> - **1:1 (One-to-One):** Each row in Entity A corresponds to exactly one row in Entity B.
> - **1:N (One-to-Many):** A single row in Entity A can link to multiple rows in Entity B.
> - **N:M (Many-to-Many):** Multiple entries in Entity A map to multiple entries in Entity B (resolved using a junction table).

Q8. Explain the purpose of AUTO_INCREMENT.

> `AUTO_INCREMENT` automatically increments numerical primary key values upon inserting new rows, preventing key collision and avoiding manual ID management.

Q9. What is the default storage engine in MySQL?

> **InnoDB** serves as the default storage engine in MySQL, offering transaction management (ACID compliance), foreign key support, and row-level locking.

Q10. What is a composite key?

> A composite key is a primary key formed by combining two or more attributes together when a single column is insufficient to uniquely identify a row in a table.